ALGEBRA 1 - UNIT 1

Section 1.2
Inputs, Outputs, and Functions

Following a rule from what goes in to what comes out.

Mathematical idea: A function is dependable: each input gets one output.

Today we will read input-output relationships from rules, tables, graphs, and situations. We will also decide when a relationship is a function.

Learning Target

Students will interpret functions as rules that connect inputs and outputs without relying on formal domain and range language.

I can statements

- I can determine outputs from given inputs.
- I can evaluate a function rule.
- I can identify inputs and outputs in a situation.
- I can describe how a function changes values.
- I can decide whether a rule gives exactly one output for each input.

What's To Come

Later in this section, you will return to the task below. For now, do not solve it. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed.

As you annotate, ask yourself: What is the input? What is the output? Where do you see a rule? Where do you see evidence that each input has exactly one output? What information might make a relationship fail to be a function?

Future Task

A school store is testing a ticket reward system. A student says the context, equation, table, and graph below all describe the same function.

Context

A student starts with 6 bonus points and earns 4 points for each ticket turned in.

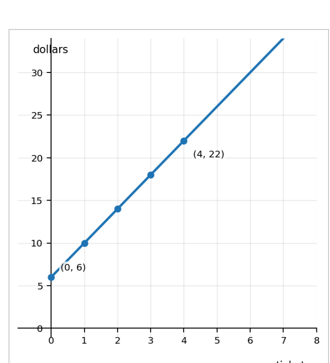
Equation

$p = 4t + 6$, where t is tickets and p is points.

Table

Tickets	0	1	2	4
Points	6	10	14	22

Graph



1. Use evidence to decide whether all four representations match.
2. Explain what the input and output mean in this situation.
3. Create one change that would make the table no longer represent a function.
4. Explain why your change breaks the idea of one output for each input.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

An **input** is the value we start with. An **output** is the value we get after a rule, situation, table, or graph does something with the input.

For example, if a machine doubles a number and then adds 1, the input 5 becomes the output 11. The input is not better or more important than the output. They work together as a pair.

A **function** is a relationship where each input has exactly one output. This does not mean every output has to be different. Two different inputs can share the same output. The problem happens when one input tries to go to two different outputs.

Functions help us make predictions. If we know the rule and the input, we can find the output. If we see a table or graph, we can look for the same idea: each input should lead to one clear answer.

Stop and Discuss

1. In your own words, what is the difference between an input and an output?
2. Why can two different inputs have the same output and still be a function?
3. What would make a relationship fail to be a function?

Inputs and Outputs Tell a Story

In a situation, the input is usually the quantity we choose or control. The output is the quantity that depends on that input.

Input-output pair: A pair of values that go together. In an ordered pair (x, y), the input is usually x and the output is usually y.

Example 1

A school club pays a one-time setup cost of $5 and then pays $3 for each poster printed.

1. Identify the input and output.
2. Find the output when the input is 4 posters.
3. Write the input-output pair.

You Try It 1

A phone charger costs $12 plus $2 for gift wrapping. Identify the input and output, then find the output if a person buys 3 chargers with gift wrapping.

Stop and Discuss

1. Which quantity did you choose in Example 1?
2. Which quantity changed because of that choice?

Function Rules Are Instructions

Function notation is another way to show an input-output rule. In $f(x)$, the x is the input. The value of $f(x)$ is the output.

Input x	→	**Rule** $2x + 5$	→	**Output** $f(x)$

Example 2

Evaluate the function rule for each input.

$$f(x) = 2x + 5$$

x	-1	0	3
$f(x)$			

You Try It 2

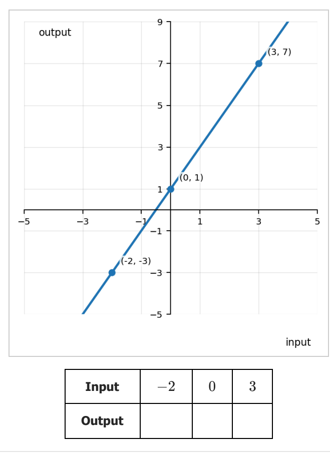
For $g(x) = 3x - 4$, find $g(-2)$, $g(1)$, and $g(5)$. Show the substitution step for at least one input.

Tables and Graphs Show the Same Input-Output Pairs

A table lists input-output pairs in rows or columns. A graph shows each pair as a point. The input tells how far left or right to go, and the output tells how far up or down to go.

Example 3

Use the graph to find the outputs for inputs -2, 0, and 3. Then record the values in the table.



Input	-2	0	3
Output			

You Try It 3

A graph includes the points $(-4, 1)$, $(0, 3)$, and $(2, 4)$. Identify the input and output in each pair, then make a table from the points.

Stop and Discuss

1. How does a table show the same information as plotted points?
2. When you read a graph, why do you need both coordinates?

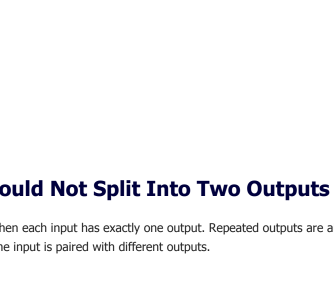
One Input Should Not Split Into Two Outputs

A relationship is a function when each input has exactly one output. Repeated outputs are allowed. Repeated inputs are only a problem when the same input is paired with different outputs.

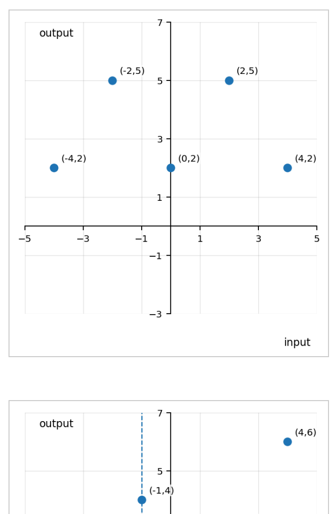
Example 4

Decide whether each relation is a function. Explain using inputs and outputs.

Relation A



Relation B



You Try It 4

A student says, "This table is not a function because the output 5 appears twice." Critique the claim.

Input	1	2	3	4
Output	5	8	5	11

Stop and Discuss

1. Why are repeated outputs allowed in a function?
2. What exact feature makes Relation B fail to be a function?

Summary

- An input is the value that goes into a rule, table, graph, or situation.
- An output is the value that comes out or depends on the input.
- Function notation like $f(x)$ means "the output when the input is x."
- A table and a graph can show the same input-output pairs.
- A function gives exactly one output for each input.
- Repeated outputs do not automatically make a relationship not a function.

Common Mistakes

- **Switching input and output:** In (x, y), the input is usually x and the output is usually y.
- **Forgetting to substitute:** To find $f(4)$, replace every x in the rule with 4.
- **Thinking repeated outputs break a function:** Different inputs may share an output.
- **Missing repeated inputs:** The same input cannot have two different outputs in a function.
- **Reading only one coordinate from a graph:** A point needs both the input and the output.

Optional Future Task Reflection

Do not solve the future task yet.

Look back at the future task. Highlight one part that makes more sense now than it did at the beginning. Circle one part that still feels unfamiliar. Write one sentence about what representation you would look at first.

Supplemental Notes

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